

PENETRATION TEST REPORT

External Reconnaissance Assessment

scanme.nmap.org — Authorized Target

Prepared by	Mark Meynisse
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Portfolio	https://markmeynisse.github.io
Target	scanme.nmap.org (45.33.32.156)
Test Date	June 25–26, 2026
Report Date	June 26, 2026
Authorization	Explicit — Nmap Project public test target
Classification	Public Portfolio Sample

CRITICAL	HIGH	MEDIUM	LOW	INFO
0	0	2	1	1

Finding Severity Summary

01

Executive Summary

This report documents an external reconnaissance assessment conducted against **scanme.nmap.org**, a host maintained by the Nmap Project and explicitly authorized for public security testing. The assessment was performed on June 25–26, 2026 by Mark Meynissse as part of an ongoing personal security research and skill development program.

The assessment followed a structured methodology: passive DNS enumeration, active port scanning, service version fingerprinting, vulnerability research, HTTP header analysis, and banner grabbing. Two medium-severity findings were identified, both related to significantly outdated and end-of-life software with publicly known CVEs.

Key findings: The target is running OpenSSH 6.6.1p1, a version affected by a username enumeration vulnerability (CVE-2018-15473) impacting versions 2.3 through 7.7. Additionally, Apache HTTP Server 2.4.7 — released in 2013 and past end-of-life — is actively disclosing its version string via response headers, facilitating targeted exploit selection by potential attackers.

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Scope & Methodology

2.1 Scope

Target Host	scanme.nmap.org
IPv4 Address	45.33.32.156
IPv6 Address	2600:3c01::f03c:91ff:fe18:bb2f
Authorization	Public – Nmap Project explicit permission
Test Type	Black-box external reconnaissance
Tools Restricted	No exploitation attempted – recon and research only

2.2 Methodology

Phase 1	DNS Enumeration	dig — resolved A records, confirmed dual-IP routing
Phase 2	Port Scanning	nmap -sV -sC — TCP SYN scan, service/version detection
Phase 3	Service Fingerprinting	Banner analysis, SSH hostkey enumeration
Phase 4	Vulnerability Research	searchsploit — CVE lookup against detected versions
Phase 5	Header Analysis	curl -I — HTTP response header inspection
Phase 6	Banner Grabbing	nc — raw TCP connection to port 31337

03 Open Ports & Services

The following ports were identified as open during the Nmap service version scan. All filtered ports (996 total) returned no-response, consistent with firewall filtering.

Port	State	Service	Version Detected	Severity
22/tcp	OPEN	ssh	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13	MEDIUM
80/tcp	OPEN	http	Apache httpd 2.4.7 (Ubuntu)	MEDIUM
9929/tcp	OPEN	nping-echo	Nping echo service	INFO
31337/tcp	OPEN	tcpwrapped	No banner – connection closed	LOW

Nmap Output (excerpt)

```
$ nmap -sV -sC scanme.nmap.org
PORT STATE SERVICE VERSION
22/tcp open  ssh OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13
80/tcp open  http Apache httpd 2.4.7 ((Ubuntu))
9929/tcp open  nping-echo Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
Nmap done: 1 IP address (1 host up) scanned in 81.04 seconds
```

04

Findings

F-001

Outdated OpenSSH 6.6.1p1 — Username Enumeration Risk

MEDIUM

CVE Reference	CVE-2018-15473 (Username Enumeration, affects 2.3-7.7) CVE-2023-38408 (agent forwarding RCE, affects <9.3p2)
CVSS Score	5.3 (Medium) — CVE-2018-15473
Affected Port	22/tcp
Current Version	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13
Current Stable	OpenSSH 9.x

Description

The target is running OpenSSH version 6.6.1p1, which is significantly behind the current stable release (9.x) and has not received security patches since its EOL. The username enumeration vulnerability (CVE-2018-15473) allows an unauthenticated attacker to determine whether a given username exists on the system by observing differential timing and response behavior during SSH authentication attempts. This information can be leveraged to support credential stuffing or brute-force attacks. searchsploit confirmed public exploit code is available for this vulnerability (linux/remote/45233.py, linux/remote/45939.py).

Evidence

```
$ searchsploit openssh 6.6 OpenSSH 2.3 < 7.7 - Username Enumeration | linux/remote/45233.py OpenSSH 2.3 < 7.7 - Username Enumeration (PoC) | linux/remote/45210.py OpenSSH < 6.6 SFTP (x64) - Command Execution | linux_x86-64/remote/45000.c
```

Recommendation

Upgrade OpenSSH to the current stable version (9.x). Enforce key-based authentication and disable password authentication. Disable root login via SSH (PermitRootLogin no). Implement fail2ban or equivalent to rate-limit authentication attempts.

F-002

EOL Apache 2.4.7 with Version Header Disclosure

MEDIUM

Description

Apache HTTP Server 2.4.7 was released in 2013 and is past end-of-life with no further security patches. The Server response header actively discloses the exact version string (Apache/2.4.7 (Ubuntu)), enabling an attacker to rapidly identify applicable CVEs and exploit code without additional reconnaissance. Numerous CVEs exist for this version range.

Evidence

```
$ curl -I http://scanme.nmap.org HTTP/1.1 200 OK Server: Apache/2.4.7 (Ubuntu) <-- version disclosed
in response header Date: Fri, 26 Jun 2026 04:45:06 GMT
```

Recommendation

Upgrade Apache to the current stable 2.4.x branch. Suppress version disclosure by setting ServerTokens Prod and ServerSignature Off in the Apache configuration. Conduct a full audit of installed modules and disable unused ones.

F-003

Port 31337 — tcpwrapped, No Banner Returned

LOW

Description

Port 31337 ("elite") accepted a TCP connection but returned no banner before closing. Nmap classified this as tcpwrapped, indicating the service completes the TCP handshake but does not respond to the probe. On this authorized test target this is intentional by the Nmap Project. In a real engagement, a non-responsive port on 31337 would warrant deeper investigation due to its historical association with backdoors and RATs.

Evidence

```
$ nc scanme.nmap.org 31337 [connection established - no banner returned - connection closed by remote
host]
```

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Analyst Notes & Next Steps

This assessment was conducted as black-box external reconnaissance only. No exploitation was attempted. The following next steps would apply in a full authorized penetration test engagement:

01	SSH Username Enumeration PoC Execute CVE-2018-15473 PoC (45233.py) against port 22 to validate the vulnerability and enumerate valid system usernames.
02	Nikto Web Scan Run nikto -h http://scanme.nmap.org against port 80 to enumerate web application vulnerabilities, misconfigurations, and outdated components.
03	OS Fingerprinting Re-run Nmap with sudo to enable -O flag for operating system detection and refine the attack surface understanding.
04	Port 31337 Deep Dive Use additional probing tools (ncat, hping3) with custom payloads to attempt banner extraction from the tcpwrapped service.
05	SSL/TLS Assessment Run testssl.sh or ssllscan against any HTTPS services to identify weak cipher suites, protocol downgrade vulnerabilities, and certificate issues.

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About the Analyst

Mark Meynisse

Self-taught cybersecurity practitioner based in Fresno, CA. Hands-on experience in penetration testing, incident response, and compliance since 2020. Completed TryHackMe Jr. Penetration Tester and SOC Analyst Level 1 learning paths. Actively pursuing CompTIA Security+ certification.

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This report was produced for portfolio and professional development purposes. scanme.nmap.org is an authorized public test target maintained by the Nmap Project. No systems were harmed and no unauthorized access was attempted.